# Jay Hemnani

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Gujarat, India (Open to Relocate)

## Education

**Pandit Deendayal Energy University (PDEU)** 2018 – 2022 | Gandhinagar, India
*B.Tech in Computer Engineering | GPA: 8.7/10*

- **Coursework:** NLP, Computer Vision, Big Data Analytics, Operating Systems, Database Systems.
- **Publications:** 2 IEEE papers (AIMV 2021): Diabetes Prediction using Stacking Classifier, Efficient Vaccine Scheduler Based on CPU Scheduling Algorithms.

## Technical Skills

**Programming:** Python, JavaScript/TypeScript, SQL, Java, Go
**AI/LLM:** LangChain, LangGraph, RAG Pipelines, Prompt Engineering, Agent Orchestration, Conversational AI
**Backend & APIs:** Node.js, Express, FastAPI, Flask, REST APIs, GraphQL, gRPC
**Databases:** PostgreSQL, MongoDB, Redis, TimescaleDB, Pinecone, ChromaDB, FAISS
**Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Kafka, Airflow
**ML/Data:** PyTorch, TensorFlow, Scikit-learn, Pandas, Hugging Face, MLflow, W&B

## Experience

**Data Analyst**, *Elite Hotel Group* Summer 2025 | Remote

- Defined **metrics and SLAs** with finance and operations; they became the spec for the ETL and forecasting work.
- Built **automated ETL pipelines** in SQL and Python, cutting manual data preparation by **40%** across multi-property analytics.
- Built **dashboards** in Tableau and Power BI for occupancy, revenue, and KPI tracking across the hotel properties.

**Technical Consultant**, *Independent* 2022 – 2024 | Remote

- Did **data analytics and pipeline consulting** for small businesses; built their reporting automation and data infrastructure.
- Designed **A/B testing frameworks** and tracked engagement metrics to improve client campaigns and product decisions.

**AI/ML Intern**, *Amnex* Jan – May 2022 | Gujarat, India

- Built a **credit fraud detection ensemble** (Random Forest, XGBoost) with SMOTE for class imbalance; **94% precision** on transaction data.
- Built **automated analytics dashboards** on statistical models for operational reporting and anomaly detection.

**SWE Intern**, *Cactus Creatives* May – Nov 2019 | Gujarat, India

- Built a **cloud-native communication platform** for first responders (microservices on Azure, REST APIs); set up **CI/CD pipelines** that cut deployment time by **60%**.

## Projects

**CAG Deep Research** | *Python, LangGraph, LangChain, Pydantic, Ollama, Groq*

- Built a **LangGraph research engine** that turns a question into a causal graph; red-team and blue-team agents test each edge in parallel and a judge rules VERIFIED, FALSIFIED or UNCLEAR.
- Search and LLM providers sit behind ports, so **7 backends** swap by config and it runs on free ones. Depth caps and loop detection keep the agent loop from running forever.

**Kayak Travel Metasearch** (team) | *Node.js, React, FastAPI, MySQL, MongoDB, Redis, Kafka, Docker*

- **3-tier system**: API gateway (JWT auth, rate limiting) in front of user, search and booking services; MySQL for bookings, MongoDB for analytics, Redis for search caching.
- My part was the back end and AI layer: a **semantic cache** that reuses LLM answers for similar queries, intent parsing, and Kafka with an in-memory stand-in for local dev.

**Stock Data Platform** | *Python, Kafka, Airflow, TimescaleDB, Chart.js, Docker*

- **Market data warehouse**: 10 tickers, full price history, 4 data sources. **23 Airflow DAGs**, a **10-table star schema**, 7 Docker services.
- Kafka streams live prices every 15s in market hours into TimescaleDB. Chart.js dashboard; one command deploys it all.

**FIFA Soccer DS** | *Python, YOLOv8, ByteTrack, FastAPI, MLflow, DVC*

- Detect, track and graph pipeline for soccer video: **22 FPS** on La Liga footage on an RTX 3070 laptop. FastAPI with a live RTSP path, MLflow, a DVC stage; GNN tactics model wired, not yet trained.

## Open Source Contributions

**360+ merged pull requests across 56 open-source projects**, including vLLM, Effect, Caddy, Lightpanda, radare2, the MCP Python SDK, DSPy, Home Assistant and NestJS.
A plain-English note on each one: jayhemnani.in/oss

## Additional

**Languages:** English, Hindi, Gujarati
**Interests:** Rubik's Cube Speedcubing (16.7s, WCA), Chess, Table Tennis, Graphic Design